



The finite element method for the coupled Schrödinger–KdV equations

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ARTICLE INFO

Article history:

Received 23 November 2008
Received in revised form 22 March 2009
Accepted 20 April 2009
Available online 24 April 2009
Communicated by A.R. Bishop

MSC:

65M60
65N15
65N30
35Q55

Keywords:

Coupled Schrödinger–KdV equations
Semi-discrete scheme
Finite element method
Conservation properties

ABSTRACT

In this Letter, we employ finite element method to study a periodic initial value problem for the coupled Schrödinger–KdV equations. For the case of one dimension, this problem is reduced to a system of ordinary differential equations by using a semi-discrete scheme. The conservation properties of this scheme, the existence and uniqueness of the discrete solutions, and error estimates are presented. In numerical experiments, the resulting system of ordinary differential equations are solved by Runge–Kutta method at each time level. The superior accuracy of this scheme is shown by comparing the numerical solutions with the exact solutions.

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1. Introduction

Nonlinear phenomena play a crucial role in a variety of scientific fields, especially in fluid mechanics, solid state physics, plasma physics, plasma waves and chemical physics. The coupled Schrödinger–KdV equations [1],

$$i\epsilon E_t + \frac{3}{2}E_{xx} - \frac{1}{2}NE = 0, \quad (1.1)$$

$$N_t + \frac{1}{2}N_{xxx} + \frac{1}{2}(N^2 + |E|^2)_x = 0, \quad (1.2)$$

have been used extensively to model nonlinear dynamics of one-dimensional Langmuir and ion-acoustic waves in a system of coordinates moving at the ion-acoustic speed. Here E is complex function describing electric field of Langmuir oscillations and N is real function describing low-frequency density perturbation. $\epsilon > 0$ is a constant. The periodic initial value problem for (1.1) and (1.2) have the following invariants:

the number of plasmons:

$$I_1 = \int_{\Omega} |E|^2 dx, \quad (1.3a)$$

the number of particles:

$$I_2 = \int_{\Omega} N dx, \quad (1.3b)$$

the energy of the oscillations:

$$I_3 = \int_{\Omega} \left[3|E_x|^2 + N|E|^2 + \frac{1}{3}N^3 - \frac{1}{2}(N_x)^2 \right] dx, \quad (1.3c)$$

which plays an important role in the analysis of this equations.

This problem has been the subject of research for many years and has motivated a series of work in physics and mathematics. The study on soliton solution has also been presented in [1–5]. The stability properties of such solutions for external perturbations as well as during interactions among themselves have been studied by a number of authors both numerically [1,6] and analytically [7–11]. For example, numerical simulations of the propagation and interactions of one-dimensional Langmuir solitons and their generation from random fluctuations by an external pump field are presented [6]. Results on existence, uniqueness of the solutions and global attractor of this systems have been obtained in [12,13]. Recently, the various techniques such as decomposition method [14], variational iteration method [15] and complex ansatz method [16] have been widely used to solve this problem. However, to the best of our knowledge, only Appert and Vaclavik [1] did some numerical work concerning this coupled Schrödinger–KdV equations

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by using finite difference Crank–Nicolson (CN) scheme, while the intensive analysis of the precise of this method is very limited. Therefore, it is of great interests to develop an efficient and accurate method for these equations.

The present Letter is an attempt to study the evolution of one-dimensional coupled Langmuir wave and ion-acoustic solitary wave by finite element method (FEM), which may be useful for further studying interactions between solitons. We construct a semi-discrete Galerkin FEM scheme. In addition to present the conservation properties of discrete scheme, we get existence and uniqueness of numerical solutions and error estimates. The space variable is discretized using quadratic splines element approximation. The system of nonlinear ordinary differential equation (ODE) which derived from the semi-discrete scheme is solved by Runge–Kutta method. We compare the solutions by our method and the CN method with exact solutions to show the accuracy and efficiency of our scheme.

This Letter is organized as follows. In Section 2, we introduce the periodic initial value problem about Schrödinger–KdV equations and some preliminaries. Section 3 is devoted to present semi-discrete scheme. Special attention is given to verify two aspects. One is that discrete scheme preserved three desirable invariants. The other is the existence, uniqueness of numerical solutions. In Section 4, the main results about error estimates are derived. Numerical experiments and data comparison are presented in Section 5. The last section is conclusion.

2. Periodic initial value problem and some preliminaries

We consider the one-dimensional problem

$$i\epsilon \frac{\partial E}{\partial t} + \frac{3}{2} \frac{\partial^2 E}{\partial x^2} - \frac{N}{2} E = 0, \quad (x, t) \in \mathbb{R} \times (0, T], \quad (2.1)$$

$$\frac{\partial N}{\partial t} + \frac{1}{2} \frac{\partial^3 N}{\partial x^3} + \frac{1}{2} (N^2 + |E|^2)_x = 0, \quad (x, t) \in \mathbb{R} \times (0, T], \quad (2.2)$$

$$E(x, t) = E(x + l, t), \quad N(x, t) = N(x + l, t),$$

$$(x, t) \in \mathbb{R} \times (0, T], \quad (2.3)$$

$$E(x, 0) = E_0(x), \quad N(x, 0) = N_0(x), \quad x \in \mathbb{R}, \quad (2.4)$$

where $E_0(x)$ and $N_0(x)$ are known l -periodic functions, $l > 0$. For the existence of solutions of (2.1)–(2.4), the proof is presented in [13, Theorem 2.1]. Furthermore, if the initial value E_0 and N_0 satisfy some smooth conditions, then the maximum-norm of solutions E and N are bounded. It can be verified that the set of Eqs. (2.1)–(2.4) preserve the referred three conservation laws.

Let $\Omega = [0, l]$. We use the notation $H^r(\Omega) = W^{r,2}(\Omega)$ to denote the complex Sobolev space on Ω . The additional of the subscript p indicates periodicity:

$$H_p^r = \{u \in H^r(\Omega) \mid \partial_x^j u(x)/\partial x^j = \partial_x^j u(x+l)/\partial x^j, j = 0, 1, \dots, r-1\},$$

where $\partial_x^j u/\partial x^j$ represent the j th derivative with respect to x of function u .

Denote $(\cdot)$ the inner product in space $L_2(\Omega)$, $\|\cdot\|$ the norm in space $L_2(\Omega)$ and $\|\cdot\|_r$ the norm in space $H^r(\Omega)$,

$$(u, v) = \int_{\Omega} u \cdot \bar{v} dx, \quad \|v\| = (v, v)^{\frac{1}{2}} = \left(\int_{\Omega} v \bar{v} dx \right)^{\frac{1}{2}},$$

$$\|v\|_r = \|v\|_{H^r} = \left(\sum_{|\alpha| \leq r} \|D^\alpha v\|^2 \right)^{\frac{1}{2}},$$

where $\bar{v}$ denotes the complex conjugate of v .

The solution space of problem (2.1)–(2.4) is defined as follows:

$$W = \left\{ u \in L^\infty(H_p^{r+1}(\Omega)), \frac{\partial u}{\partial t} \in L^\infty(H_p^{r+1}(\Omega)), r \geq 3 \right\}.$$

Now we write the problem (2.1)–(2.4) in weak form: find $E, N \in H_p^r(\Omega)$, such that

$$i\epsilon \left(\frac{\partial E}{\partial t}, \psi \right) - \frac{3}{2} \left(\frac{\partial E}{\partial x}, \frac{\partial \psi}{\partial x} \right) - \frac{1}{2} (NE, \psi) = 0, \quad (2.5)$$

$$\left(\frac{\partial N}{\partial t}, \psi \right) - \frac{1}{2} \left(\frac{\partial^2 N}{\partial x^2}, \frac{\partial \psi}{\partial x} \right) - \frac{1}{2} (N^2 + |E|^2, \frac{\partial \psi}{\partial x}) = 0, \quad (2.6)$$

$$E(x, 0) = E_0(x), \quad N(x, 0) = N_0(x), \quad (2.7)$$

where $\psi \in H_p^2(\Omega)$.

3. Semi-discrete scheme

Let M be a positive integer. Define the spatial mesh size $h = l/M$, and the grid points $x_j = jh$, $j = 0, 1, \dots, M$. $S_h(\Omega)$ denote the space of l -periodic piecewise polynomial functions of degree $r-1$ on subinterval $[x_j, x_{j+1}]$. $S_h(\Omega)$ possesses the well-known approximation properties [17,18]:

If $u \in H^r(\Omega) \cap H_0^1(\Omega)$, then

$$\inf_{\chi \in S_h} \{ \|u - \chi\| + h \|u - \chi\|_1 \} \leq Ch^s \|u\|_s, \quad 1 \leq s \leq r, \quad (3.1)$$

and, for all $\chi \in S_h$, the following inverse inequality holds:

$$\|\chi\|_\beta \leq Ch^{-(\beta-\alpha)} \|\chi\|_\alpha, \quad 0 \leq \alpha \leq \beta \leq r-1, \quad (3.2)$$

where C is independent of h .

Our semi-discrete approximation problem is defined as: find $E_h = E_h(x, t)$ and $N_h = N_h(x, t)$ belong to S_h , such that

$$i\epsilon \left(\frac{\partial E_h}{\partial t}, \psi \right) - \frac{3}{2} \left(\frac{\partial E_h}{\partial x}, \frac{\partial \psi}{\partial x} \right) - \frac{1}{2} (N_h E_h, \psi) = 0, \quad (3.3)$$

$$\left(\frac{\partial N_h}{\partial t}, \psi \right) - \frac{1}{2} \left(\frac{\partial^2 N_h}{\partial x^2}, \frac{\partial \psi}{\partial x} \right) - \frac{1}{2} (N_h^2 + |E_h|^2, \frac{\partial \psi}{\partial x}) = 0, \quad (3.4)$$

$$E_h(x, 0) = E_{0h}, \quad N_h(x, 0) = N_{0h}, \quad (3.5)$$

where $\psi \in S_h$, E_{0h} and N_{0h} are some approximation to E_0 and N_0 in S_h , respectively.

Theorem 1. Let E_h and N_h be the solution of (3.3)–(3.5), then the following conservation properties hold:

$$I_1^* = \int_{\Omega} |E_h|^2 dx = \int_{\Omega} |E_{0h}|^2 dx, \quad (3.6)$$

$$I_2^* = \int_{\Omega} N_h dx = \int_{\Omega} N_{0h} dx, \quad (3.7)$$

$$\begin{aligned} I_3^* &= \int_{\Omega} \left[3|E_{hx}|^2 + N_h |E_h|^2 + \frac{1}{3} N_h^3 - \frac{1}{2} (N_{hx})^2 \right] dx \\ &= \int_{\Omega} \left[3|E_{0hx}|^2 + N_{0h} |E_{0h}|^2 + \frac{1}{3} N_{0h}^3 - \frac{1}{2} (N_{0hx})^2 \right] dx. \end{aligned} \quad (3.8)$$

Proof. Let $\psi = E_h$ in (3.3), we have

$$i\epsilon \left(\frac{\partial E_h}{\partial t}, E_h \right) - \frac{3}{2} \left(\frac{\partial E_h}{\partial x}, \frac{\partial E_h}{\partial x} \right) - \frac{1}{2} (N_h E_h, E_h) = 0. \quad (3.9)$$

According to the definition of inner product, the (3.9) is equal to

$$i\epsilon \int_{\Omega} \frac{\partial E_h}{\partial t} \cdot \bar{E}_h dx - \frac{3}{2} \int_{\Omega} \frac{\partial E_h}{\partial x} \cdot \frac{\partial \bar{E}_h}{\partial x} dx - \frac{1}{2} \int_{\Omega} N_h \cdot E_h \cdot \bar{E}_h dx = 0, \quad (3.10)$$

or

$$i\epsilon \int_{\Omega} \frac{\partial E_h}{\partial t} \cdot \bar{E}_h dx - \frac{3}{2} \int_{\Omega} \left| \frac{\partial E_h}{\partial x} \right|^2 dx - \frac{1}{2} \int_{\Omega} N_h \cdot |E_h|^2 dx = 0. \quad (3.11)$$

Then calculating the conjugate of the above ODE (3.11), we have

$$-i\epsilon \int_{\Omega} \frac{\partial \bar{E}_h}{\partial t} \cdot E_h dx - \frac{3}{2} \int_{\Omega} \left| \frac{\partial E_h}{\partial x} \right|^2 dx - \frac{1}{2} \int_{\Omega} N_h \cdot |E_h|^2 dx = 0. \quad (3.12)$$

Subtracting (3.12) from (3.11), one gets

$$i\epsilon \int_{\Omega} \left(\frac{\partial E_h}{\partial t} \cdot \bar{E}_h + \frac{\partial \bar{E}_h}{\partial t} \cdot E_h \right) dx = 0$$

or

$$\int_{\Omega} \frac{d}{dt} |E_h|^2 dx = 0.$$

The derivative action is in the direction of time, while the integral action is in the direction of space. So the process of differential and integral are exchangeable. Therefore, we have

$$\frac{d}{dt} \int_{\Omega} |E_h|^2 dx = 0.$$

It is easy to obtain (3.6). One note that (3.4) is equal to

$$\left(\frac{\partial N_h}{\partial t}, \psi \right) + \frac{1}{2} \left(\frac{\partial^3 N_h}{\partial x^3}, \psi \right) + \frac{1}{2} ((N_h^2 + |E_h|^2)_x, \psi) = 0.$$

Setting $\psi = 1$ in it, by periodicity we have

$$\frac{d}{dt} \int_{\Omega} N_h dx = 0.$$

This implies that (3.7) holds. To prove (3.8), we choose $\psi = E_{ht}$ in (3.3) and take real part. This yields that

$$3 \frac{d}{dt} \int_{\Omega} |E_{hx}|^2 dx + \frac{d}{dt} \int_{\Omega} N_h |E_h|^2 dx - \int_{\Omega} N_{ht} |E_h|^2 dx = 0. \quad (3.13)$$

Subsequently, setting $\psi = |E_h|^2$, N_h^2 and $\frac{\partial^2 N_h}{\partial x^2}$ in (3.4) respectively, by periodicity we obtain

$$\int_{\Omega} N_{ht} |E_h|^2 dx - \frac{1}{2} \int_{\Omega} \frac{\partial^2 N_h}{\partial x^2} (|E_h|^2)_x dx - \frac{1}{2} \int_{\Omega} N_h^2 (|E_h|^2)_x dx = 0, \quad (3.14)$$

$$\frac{d}{dt} \int_{\Omega} N_h^3 dx - \frac{1}{2} \int_{\Omega} \frac{\partial^2 N_h}{\partial x^2} (N_h^2)_x dx - \frac{1}{2} \int_{\Omega} |E_h|^2 (N_h^2)_x dx = 0, \quad (3.15)$$

and

$$-\frac{1}{2} \frac{d}{dt} \int_{\Omega} \left(\frac{\partial N_h}{\partial x} \right)^2 dx + \frac{1}{2} \int_{\Omega} \frac{\partial^2 N_h}{\partial x^2} (N_h^2 + |E_h|^2)_x dx = 0. \quad (3.16)$$

Combining (3.14)–(3.16) yields

$$\frac{d}{dt} \int_{\Omega} \left(3|E_{hx}|^2 + N_h |E_h|^2 + \frac{1}{3} N_h^3 - \frac{1}{2} (N_{hx})^2 \right) dx = 0. \quad (3.17)$$

This completes the proof of (3.8). $\square$

4. Existence, uniqueness and error estimates

Theorem 2. There exists unique E_h and N_h satisfying (3.3)–(3.5).

Proof. Let $\{\phi_i\}_{i=1}^M$ be a basis $S_h(\Omega)$. The approximate solutions E_h and N_h can be expressed as

$$E_h = \sum_{i=1}^M E_{hi}(t) \phi_i(x), \quad N_h = \sum_{i=1}^M N_{hi}(t) \phi_i(x), \quad (4.1)$$

where the coefficients $\{E_{hi}(t)\}_{i=1}^M$ and $\{N_{hi}(t)\}_{i=1}^M$ are the complex and real functions of t , respectively. Using this, (3.3)–(3.4) are reduced to an initial value problem of a system of ODE: Find $E_{hi}(t)$ and $N_{hi}(t)$ such that

$$i\epsilon \sum_{i=1}^M E'_{hi}(t) (\phi_i(x), \phi_j(x)) - \frac{3}{2} \sum_{i=1}^M E_{hi}(t) (\phi_{ix}(x), \phi_{jx}(x)) - \frac{1}{2} \left(\left(\sum_{i=1}^M N_{hi}(t) \phi_i(x) \right) \cdot \left(\sum_{i=1}^M E_{hi}(t) \phi_i(x) \right), \phi_j(x) \right) = 0, \quad (4.2)$$

$$\sum_{i=1}^M N'_{hi}(t) (\phi_i(x), \phi_j(x)) - \frac{1}{2} \sum_{i=1}^M N_{hi}(t) (\phi_{ixx}(x), \phi_{jx}(x)) - \frac{1}{2} \left(\left(\sum_{i=1}^M N_{hi}(t) \phi_i(x) \right)^2 + \left(\sum_{i=1}^M E_{hi}(t) \phi_i(x) \right)^2, \phi_{jx}(x) \right) = 0, \quad (4.3)$$

$$E_{hj}(0) = E_{0h}(x_j), \quad N_{hj}(0) = N_{0h}(x_j), \quad (4.4)$$

where $j = 1, 2, \dots, M$.

Let $(E_{h1}(t), E_{h2}(t), \dots, E_{hM}(t))^T = \bar{Y}_1(t)$ and $(N_{h1}(t), N_{h2}(t), \dots, N_{hM}(t))^T = \bar{Y}_2(t)$ are two M -dimensional vector functions. Then in matrix notation (4.2)–(4.3) are reduced to

$$i\epsilon A \frac{d\bar{Y}_1}{dt} = \frac{3}{2} B \bar{Y}_1 + \frac{1}{2} C \bar{Y}_1, \quad (4.5)$$

$$A \frac{d\bar{Y}_2}{dt} = \frac{1}{2} D \bar{Y}_2 + \frac{1}{2} G_1 \bar{Y}_1 + \frac{1}{2} G_2 \bar{Y}_2, \quad (4.6)$$

with initial conditions $\bar{Y}_1(0) = (E_{0h}(x_1), E_{0h}(x_2), \dots, E_{0h}(x_M))^T$ and $\bar{Y}_2(0) = (N_{0h}(x_1), N_{0h}(x_2), \dots, N_{0h}(x_M))^T$. Here A , B and D are $M \times M$ matrices whose entries are given by

$$A_{ij} = (\phi_i, \phi_j), \quad B_{ij} = (\phi_{ix}, \phi_{jx}), \quad D_{ij} = (\phi_{ixx}, \phi_{jx}),$$

for $1 \leq i, j \leq M$. $C \bar{Y}_1$, $G_1 \bar{Y}_1$, $G_2 \bar{Y}_2$ are M -dimensional vector whose components are

$$(C \bar{Y}_1)_i = \sum_{k,j=1}^M E_{hk} N_{hj} (\phi_k \phi_j, \phi_i),$$

$$(G_1 \bar{Y}_1)_i = \sum_{k,j=1}^M E_{hk} \bar{E}_{hj} (\phi_k \phi_j, \phi_{ix}),$$

$$(G_2 \bar{Y}_2)_i = \sum_{k,j=1}^M N_{hk} N_{hj} (\phi_k \phi_j, \phi_{ix}),$$

for $1 \leq i \leq M$. The mass matrix A is Gram matrix and invertible, so that the system of ODE (4.5) and (4.6) can be written as

$$i\epsilon \frac{d\bar{Y}'_1}{dt} = \frac{3}{2} A^{-1} B \bar{Y}_1 + \frac{1}{2} A^{-1} C \bar{Y}_1, \quad (4.7)$$

$$\frac{d\bar{Y}'_2}{dt} = \frac{1}{2} A^{-1} D \bar{Y}_2 + \frac{1}{2} A^{-1} G_1 \bar{Y}_1 + \frac{1}{2} A^{-1} G_2 \bar{Y}_2. \quad (4.8)$$

Observe the every terms of the right sides of (4.7) and (4.8), simply denote them with $F_1(\bar{Y}_1, \bar{Y}_2)$ and $F_2(\bar{Y}_1, \bar{Y}_2)$. Because of the characteristic of finite element solutions, we know that $\|E_h\|_\infty$ and $\|N_h\|_\infty$ are bounded, thus for a fixed x_i , $\|E_{hi}(t)\|$ and $\|N_{hi}(t)\|$ are bounded too. It can be proved easily that F_1 and F_2 are Lipschitz continuous with respect to $\bar{Y}_1(t)$ and $\bar{Y}_2(t)$. Therefore, the system of ODE (4.7) and (4.8) with initial value $\bar{Y}_1(0)$ and $\bar{Y}_2(0)$ exist unique solution. The proof of existence and uniqueness is complete. $\square$

Theorem 3. Assume that E and N belong to $H_p^{r+1}(\Omega)$ and are the solutions of (2.1)–(2.4). If E_h and N_h are the solutions of (3.3)–(3.5) in $S_h(\Omega)$, then there has a constant C depending only on E , N and T , such that

$$\|N_h - N\| \leq Ch^r,$$

$$\|E_h - E\| \leq Ch^r.$$

In order to estimate the error, we firstly introduce the auxiliary projection P_h on $S_h(\Omega)$ defined by: for $v \in H_p^2(\Omega)$

$$(\nabla(P_h v - v), \nabla \chi) = 0, \quad \forall \chi \in S_h, \quad (4.9)$$

where P_h satisfies the following inequalities:

Lemma 2. (See [17,18].) Assuming that a function v satisfies conditions:

$$v \in L^\infty(H^{r+1}(\Omega)), \quad \frac{\partial v}{\partial t} \in L^\infty(H^{r+1}(\Omega)),$$

then there holds the following inequalities:

$$\|P_h v - v\| + h \|\nabla(P_h v - v)\| \leq M_1 h^{r+1} \|v\|_{r+1}, \quad (4.10)$$

$$\left\| \frac{\partial(P_h v - v)}{\partial t} \right\| + h \left\| \nabla \left(\frac{\partial(P_h v - v)}{\partial t} \right) \right\| \leq M_2 h^{r+1} \left\| \frac{\partial v}{\partial t} \right\|_{r+1}, \quad (4.11)$$

where M_1, M_2 are the constants independent of h .

Before proving this theorem, we give the following lemma.

Lemma 3. ([19], Continuous Gronwall inequality). Suppose that the continuous function $\eta(t)$ satisfies

$$|\eta(t)| \leq \beta + \alpha \int_a^t |\eta(\tau)| d\tau, \quad a \leq t \leq b,$$

where α, β are nonnegative constant. Then

$$|\eta(t)| \leq \beta e^{\alpha(t-a)}, \quad a \leq t \leq b.$$

Our approach is based on the error decomposition with

$$E_h - E = (E_h - P_h E) + (P_h E - E) = \theta_1 + \rho_1, \quad (4.12)$$

$$N_h - N = (N_h - P_h N) + (P_h N - N) = \theta_2 + \rho_2. \quad (4.13)$$

Proof of Theorem 3. It is shown that ρ_1 and ρ_2 , the second terms on the right side of (4.12) and (4.13), are easily bounded by inequalities (4.10). Therefore our attention is given to estimate θ_1 and θ_2 . By periodicity we get the following error equations

$$i\epsilon \left(\frac{\partial \theta_1}{\partial t}, \psi \right) + i\epsilon \left(\frac{\partial \rho_1}{\partial t}, \psi \right) - \frac{3}{2} \left(\frac{\partial \theta_1}{\partial x}, \frac{\partial \psi}{\partial x} \right) - \frac{1}{2} (N_h E_h - N E, \psi) = 0, \quad (4.14)$$

$$\left(\frac{\partial \theta_2}{\partial t}, \psi \right) + \left(\frac{\partial \rho_2}{\partial t}, \psi \right) - \frac{1}{2} \left(\frac{\partial^2 \theta_2}{\partial x^2}, \frac{\partial \psi}{\partial x} \right) - \frac{1}{2} (N_h^2 - N^2, \frac{\partial \psi}{\partial x}) - \frac{1}{2} (|E_h|^2 - |E|^2, \frac{\partial \psi}{\partial x}) = 0. \quad (4.15)$$

Setting $\psi = \theta_1$ in (4.14) and taking the imaginary parts yields

$$\begin{aligned} \frac{d}{dt} \|\theta_1\|^2 &= -2 \operatorname{Re} \left(\frac{\partial \rho_1}{\partial t}, \theta_1 \right) + \operatorname{Im} \frac{1}{\epsilon} (N_h E_h - N E, \theta_1) \\ &\leq C \left(\|\theta_1\|^2 + \|\rho_1\|^2 + \|\theta_2\|^2 + \|\rho_2\|^2 + \left\| \frac{\partial \rho_1}{\partial t} \right\|^2 \right). \end{aligned} \quad (4.16)$$

Here we used Hölder's inequality and the boundedness of $\|E_h\|_\infty$ and $\|N_h\|_\infty$. Taking $\psi = \theta_2$ in (4.15), by periodicity we get

$$\begin{aligned} \frac{d}{dt} \|\theta_2\|^2 &= - \left(\frac{\partial \rho_2}{\partial t}, \theta_2 \right) + \frac{1}{2} (N_h^2 - N^2, \frac{\partial \theta_2}{\partial x}) + \frac{1}{2} (|E_h|^2 - |E|^2, \frac{\partial \theta_2}{\partial x}) \\ &\leq C \left(\|\theta_2\|^2 + \|\rho_2\|^2 + \|\theta_1\|^2 + \|\rho_1\|^2 + \left\| \frac{\partial \rho_2}{\partial t} \right\|^2 + \left\| \frac{\partial \theta_2}{\partial x} \right\|^2 \right). \end{aligned} \quad (4.17)$$

Noting the inverse inequality (3.2), we have $\|\frac{\partial \theta_2}{\partial x}\| \leq Ch^{-1} \|\theta_2\|$. Together with the above two inequalities (4.16) and (4.17), we get

$$\begin{aligned} \frac{d}{dt} (\|\theta_1\|^2 + \|\theta_2\|^2) &\leq Ch^{-2} \left(\|\theta_1\|^2 + \|\theta_2\|^2 + \|\rho_1\|^2 + \|\rho_2\|^2 + \left\| \frac{\partial \rho_1}{\partial t} \right\|^2 + \left\| \frac{\partial \rho_2}{\partial t} \right\|^2 \right). \end{aligned} \quad (4.18)$$

Integrating (4.18) from 0 to t yields

$$\begin{aligned} \|\theta_1(t)\|^2 + \|\theta_2(t)\|^2 &\leq \|\theta_1(x, 0)\|^2 + \|\theta_2(x, 0)\|^2 \\ &\quad + Ch^{-2} \int_0^t (\|\theta_1(\tau)\|^2 + \|\theta_2(\tau)\|^2 + \|\rho_1(\tau)\|^2 + \|\rho_2(\tau)\|^2 + \left\| \frac{\partial \rho_1}{\partial t}(\tau) \right\|^2 + \left\| \frac{\partial \rho_2}{\partial t}(\tau) \right\|^2) d\tau. \end{aligned} \quad (4.19)$$

Next, we make special choice of $E_{0h} = P_h E_0$ and $N_{0h} = P_h N_0$ so as to bound the two terms $\|\theta_1(0)\|$ and $\|\theta_2(0)\|$. Since $\|\rho_1\|$, $\|\frac{\partial \rho_1}{\partial t}\|$, $\|\rho_2\|$ and $\|\frac{\partial \rho_2}{\partial t}\|$ are bounded as claimed in Lemma 2. Thus applying Lemma 3, we have

$$\begin{aligned} \|\theta_1(t)\|^2 + \|\theta_2(t)\|^2 &\leq Ch^{2r} \int_0^t (\|E\|^2 + \|N\|^2) d\tau \\ &\quad + Ch^{-2} \int_0^t (\|\theta_1\|^2 + \|\theta_2\|^2) d\tau \\ &\leq C e^{C(t-\tau)} h^{2r} \\ &\leq C e^{CT} h^{2r}. \end{aligned} \quad (4.20)$$

Then we obtain

$$\|\theta_1\| + \|\theta_2\| \leq Ch^r.$$

Writing again $E_h - E = \theta_1 + \rho_1$, $N_h - N = \theta_2 + \rho_2$ and using the estimates for θ_1 and θ_2 , we get

$$\|N_h - N\| \leq Ch^r, \quad \|E_h - E\| \leq Ch^r.$$

This completes the proof of this theorem. $\square$

5. Numerical experiments

In this section, we present some numerical results for the proposed scheme (3.3)–(3.5). The accuracy of our finite element method is tested by looking at the error using L_2 norm and the discrete conservation properties, which are important feature of smooth soliton solution.

Eqs. (1.1) and (1.2) admit a stationary traveling localized solution, C-soliton, given by [12]:

$$E(x, t) = -\frac{6}{5}\sqrt{3}\alpha \frac{1}{\cosh \xi} \tanh \xi \times \exp\left\{i\alpha\left[\left(\frac{3}{20\epsilon} - \frac{\epsilon\alpha}{6}\right)t - \frac{\epsilon x}{3}\right]\right\}, \quad (5.1)$$

$$N(x, t) = -\frac{9}{5}\alpha \frac{1}{\cosh^2 \xi}, \quad (5.2)$$

where $\xi = (\alpha/10)^{1/2}(x + \alpha t)$, and α is a free positive parameter. We take the value of the exact solution (5.1) and (5.2) at $t = 0$ as our initial conditions:

$$E_0 = -\frac{6}{5}\alpha \frac{1}{\cosh \xi} \tanh \xi \exp\left\{i\alpha\left(-\frac{\epsilon x}{3}\right)\right\},$$

$$N_0 = -\frac{9}{5}\alpha \frac{1}{\cosh^2 \xi},$$

where $\xi = (\alpha/10)^{1/2}x$.

We choose quadratic spline functions as basis functions. The compound trapezoidal integral formula is utilized to compute the integration in coefficient matrices of (4.5) and (4.6). To test whether the proposed numerical scheme exhibit the expected error order in space, we perform some numerical experiments for different values of h and fixed time $t = 0.001$ with the time step length $\kappa = 1 \times 10^{-5}$ to ensure the temporal error is negligible. Meanwhile, fourth order Runge–Kutta method is used to solve the system of ODE at each time level.

The accuracy is measured by L_2 error norm defined by

$$e = \|u(x, t) - u_h\| = \left[h \sum_j |u(x_j, t) - u_{hj}|^2\right]^{1/2}.$$

The error norms of the imaginary part and real part of E_h and N_h are denoted with e_1, e_2 and e_3 respectively. The order of error is calculated using the following formula:

$$\text{order of error} \approx \frac{\ln(e(h))}{\ln(h)}.$$

Computations were done with parameters $\epsilon = 1$, $\alpha = 0.45$, $x_L = -30$, $x_R = 30$. Tables 1–3 and 4–6 show the numerical and exact solution with $h = 0.25$ and $h = 0.05$ at $t = 0.001$ respectively. It can be seen that the error norm calculated by the finite element method is smaller than that of the Crank–Nicolson method. Moreover, the error order of real part and imaginary part of E_h is bigger than 2.6, which is well agreement with the conclusion that the value of r in aforementioned L_2 error norm $e \approx O(h^r)$ is at least 3 because of the existing of third degree derivative term in the coupled equations. Meanwhile, the error order of N_h is larger

Table 1

Comparison of numerical with exact solutions for the image part of $E(x, t)$ at $t = 0.001$ for $h = 0.25$.

x	FEM	CN	Exact solution
−20	−3.78879e−003	−3.89860e−003	−3.79041e−003
−10	−2.14408e−001	−2.14246e−001	−2.142781e−001
0	−1.06570e−005	−3.53764e−003	−3.013331e−005
10	−2.14246e−001	−2.14250e−001	−2.142397e−001
20	−3.79314e−003	−3.89860e−003	−3.791490e−003
e_1	3.66905e−004	2.02026e−003	
order	2.6405	2.0711	

Table 2

Comparison of numerical and exact solutions for the real part of $E(x, t)$ at $t = 0.001$ for $h = 0.25$.

x	FEM	CN	Exact solution
−20	−2.66051e−002	−2.66155e−002	−2.65971e−002
−10	1.52075e−002	1.59503e−002	1.51882e−002
0	1.43760e−005	5.23673e−005	−8.92839e−005
10	−1.51723e−002	−1.59516e−002	−1.52000e−002
20	2.65836e−002	2.66155e−002	2.65918e−002
e_2	2.54068e−004	3.47759e−003	
order	2.7632	1.8898	

Table 3

Comparison of numerical and exact solutions for $N(x, t)$ at $t = 0.001$ for $h = 0.25$.

x	FEM	CN	Exact solution
−20	−6.69045e−004	−6.71879e−004	−6.68863e−004
−10	−4.52675e−002	−4.54481e−002	−4.52561e−002
0	−8.10000e−001	−8.09996e−001	−8.09999e−001
10	−4.52279e−002	−4.50475e−002	−4.52393e−002
20	−6.68427e−004	−6.65594e−004	−6.68608e−004
e_3	1.51938e−005	3.36311e−004	
order	3.7034	2.6696	

Table 4

Comparison of numerical and exact solutions for the image part of $E(x, t)$ at $t = 0.001$ for $h = 0.05$.

x	FEM	CN	Exact solution
−20	−3.80170e−003	−3.79148e−003	−3.79041e−003
−10	−2.14255e−001	−2.14239e−001	−2.142781e−001
0	−3.58321e−004	−1.04111e−008	−3.013331e−005
10	−2.14256e−001	−2.14278e−001	−2.142397e−001
20	−3.80170e−003	−3.79042e−003	−3.791490e−003
e_1	2.07954e−004	5.86211e−004	
order	2.8300	2.4841	

Table 5

Comparison of numerical and exact solutions for the real part of $E(x, t)$ at $t = 0.001$ for $h = 0.05$.

x	FEM	CN	Exact solution
−20	−2.65966e−002	−2.65918e−002	−2.65971e−002
−10	1.51982e−002	1.52695e−002	1.51882e−002
0	−8.90541e−005	−1.43760e−005	−8.92839e−005
10	−1.519006e−002	−1.52699e−002	−1.52000e−002
20	2.65966e−002	2.65971e−002	2.65918e−002
e_2	8.31830e−005	3.61245e−004	
order	3.1359	2.6457	

than 3 to 5.2956 when $h = 0.05$, which is even better than the theoretical results given in Theorem 3. In Table 7, we display the discrete conserved quantities I_1^* , I_2^* and I_3^* , obtained from the proposed scheme, which is consistent with the results of Theorem 1.

Figs. 1–4 plot the evolution of the exact and the numerical solution at different time. It can be found that our method only induces a very small numerical reflection. In particular, the curves of $|E|$ in Figs. 1 and 3 only produce very little difference with the evolution of time, and as well the curves of N in Figs. 2 and 4 al-

Table 6
Comparison of numerical and exact solutions for $N(x, t)$ at $t = 0.001$ for $h = 0.05$.

x	FEM	CN	Exact solution
−20	−6.6886389e−004	−6.6905002e−004	−6.6886378e−004
−10	−4.5256165e−002	−4.5267806e−002	−4.5256158e−002
0	−8.0999997e−001	−8.0999997e−001	−8.0999999e−001
10	−4.5239361e−002	−4.5227724e−002	−4.5239371e−002
20	−6.6860836e−004	−6.6842281e−004	−6.6860853e−004
e_3	1.288694e−007	3.382211e−005	
order	5.2956	3.4363	

Table 7
The conservative quantities of I_1^*, I_2^*, I_3^* computing with finite element method.

Time	I_1^*	I_2^*	I_3^*
0.001	2.749191	−7.636716	−1.537611
0.01	2.749191	−7.636716	−1.537611
0.1	2.749191	−7.636716	−1.537611
1	2.749191	−7.636716	−1.537611
1.5	2.749191	−7.636716	−1.537607
2	2.749192	−7.636716	−1.537601

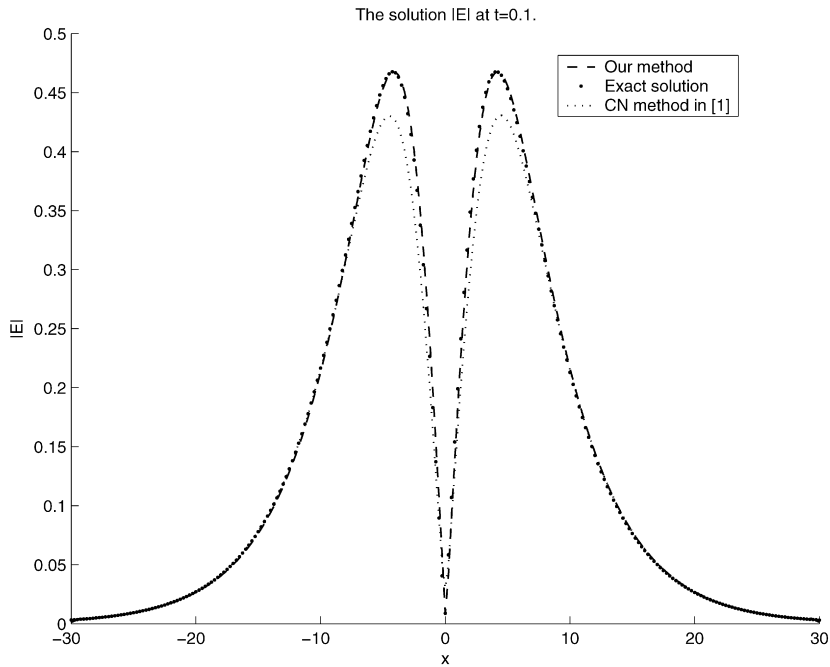


Fig. 1. The evolution of $|E_h|$ at $t = 0.1$.

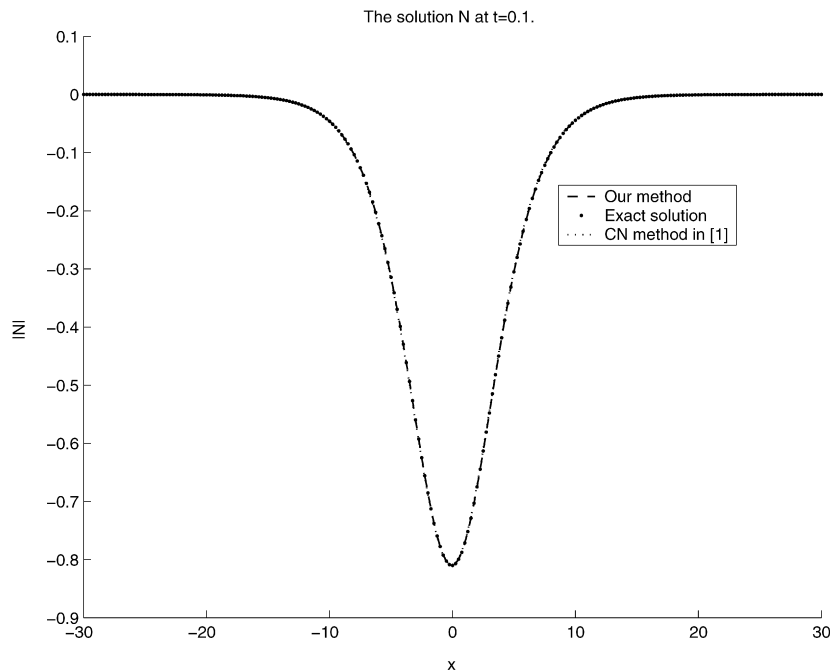
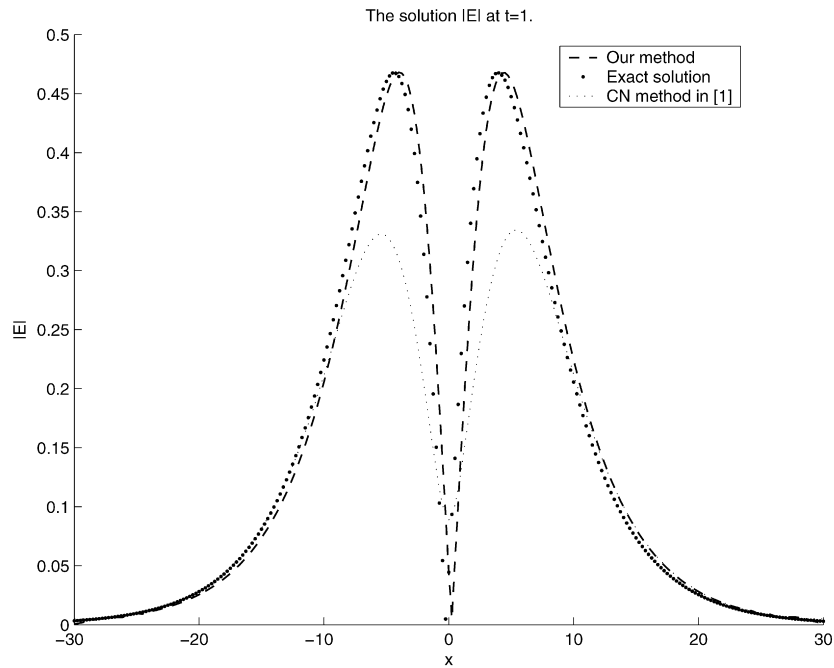
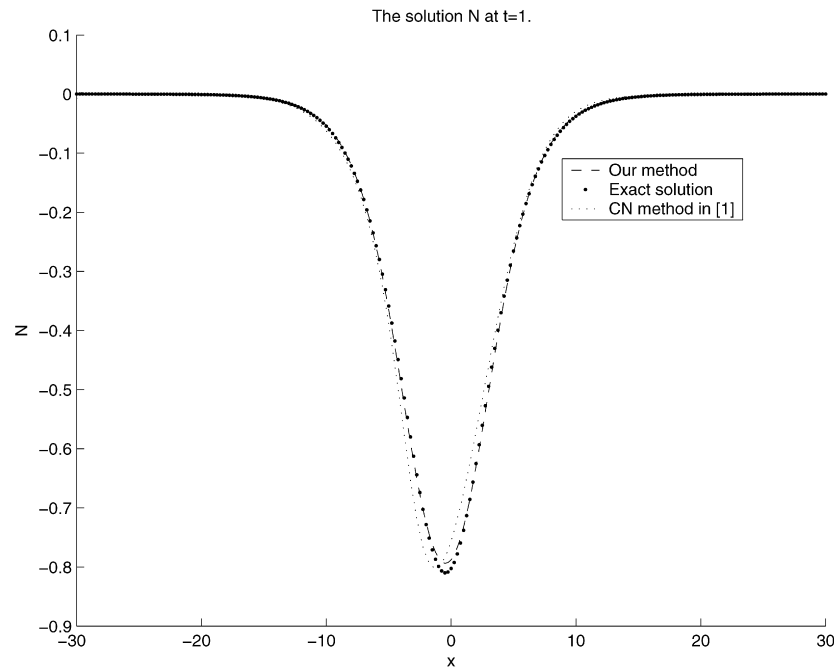


Fig. 2. The evolution of N_h at $t = 0.1$.

Fig. 3. The evolution of $|E_h|$ at $t = 1$.Fig. 4. The evolution of N_h at $t = 1$.

most coincide with the exact solution. The reason maybe due to that the error order of FEM on spacial variable is close to 3, and on temporal is 2, while the error order is 2 on both spatial and temporal variable for the CN method.

6. Conclusion

In summary, a finite element method for the coupled Schrödinger–KdV equation is presented. The quadratic elements are chosen as our basis functions. Fourth order Runge–Kutta method is applied to solve the differential equation systems which obtained from the semi-discrete scheme. Numerical experiments verify that discrete scheme preserve the three conservative quantities. The accuracy of

this scheme is better than the Crank–Nicolson method by comparing the numerical solution with the exact solution. The agreement of the error estimates between theoretical analysis and Numerical results shows that our method is efficient.

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